Luke Pitstick

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Education

University of Colorado Boulder Aug. 2024 – May 2028
Bachelor of Arts in Computer Science and Political Science *Boulder, CO*

Work Experience

WattByte Nexus Jun. 2026 – Present
AI/ML Research Intern *Golden, CO*

- Design and implement physics-informed PyTorch models to predict and reduce wildfire risk from power-grid infrastructure.
- Build event-driven wildfire-risk simulations using NOAA weather data, AWS, and Redis to model grid assets, environmental conditions, and failure scenarios.
- Design and build an AWS pipeline using Fargate, SQS, and Glue to process video footage into segmented 3D representations.

National Oceanic and Atmospheric Administration / CIRES Oct. 2025 – Present
Junior Data Manager *Boulder, CO*

- Automate ingestion, validation, and archival workflows for 10TB+/week of ocean acoustic data using Python, Oracle DB, and AWS S3.
- Maintain end-to-end sonar data workflows ingesting water column and passive acoustic data from NOAA research vessels into NCEI's national archive.
- Develop Python tools to automate QC and enforce metadata standards across oceanographic data submissions to NOAA/NCEI.

Ranked Choice Voting for Longmont Aug. 2025 – Dec. 2025
Data Science Intern *Boulder, CO*

- Engineered high-resolution demographic maps in ArcGIS, R, and Python, enabling campaign teams to identify turnout gaps and execute targeted outreach.
- Developed fixed-effects regression models to predict voter turnout, allowing the campaign team to make targeted decisions.

Leadership Experience

University of Colorado Student Government May 2025 – May 2026
Representative-at-Large & Press Secretary *Boulder, CO*

- Supported allocation of a $36M student budget across 12 cost centers, coordinating with university administrators to ensure equitable fund distribution.
- Led a 6-person team to increase student engagement across social media channels by 15%.

Projects

Swipe-Based Recipe Recommendation App | *React Native, Python, pgvector, Figma* May 2026 – Present

- Develop a mobile recipe app that helps users discover recipes, save favorites, and plan meals for the week.
- Build a personalized recipe recommendation engine using vector embeddings and pgvector to learn user preferences from swipe behavior.
- Design and develop a recipe ingestion pipeline to scrape, clean, embed, and index recipes for personalized content discovery.

InfraDrone | *Python, C++, PyTorch, TensorRT, OpenCV* | GitHub Apr. 2026 – Present

- Design and build an autonomous drone and ground station using a Jetson Orin Nano, MAVLink, and TensorRT for offline road-damage surveys.
- Trained YOLO models to detect and classify road damage, achieving 80% precision and 75% mAP@50 on validation data.
- Develop a temporal analysis engine using YOLO segmentation and geospatial alignment to estimate crack growth over time.

Renewably | *Python, XGBoost, Polars, React, FastAPI* | Demo Apr. 2026

- Won 1st place at BlasterHacks 2026 for building an interactive renewable-energy siting platform with React and ArcGIS.
- Trained an XGBoost model on geospatial features to score wind-farm site viability against existing wind turbine locations. Achieved 96% accuracy on validation data.
- Built a FastAPI optimization pipeline to evaluate thousands of candidate locations within user-defined regions.

Technical Skills

Languages: Python, TypeScript, SQL, R
Frameworks/Libraries: PyTorch, NumPy, scikit-learn, pandas, Polars, React, FastAPI
Cloud/Databases: AWS, PostgreSQL, Redis, Supabase
Other: Docker, Git, Linux, ArcGIS, Figma